

RAPID CORTEX

Stress Test Report

Production API Infrastructure Validation

Overall Verdict: PASS

Environment	PROD
API Target	https://k26yw4o3xk.execute-api.us-east-1.amazonaws.com
Generated	2026-08-27 00:45 UTC
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Classification	CONFIDENTIAL — Internal Use Only

Intelligence at the speed of response

Executive Summary

Overall Verdict

PASS

All SLA gates passed across smoke and load profiles. Ready for pilot traffic escalation.

Test Configuration

Parameter	Value
Environment / Stage	PROD
API Base URL	https://k26yw4o3xk.execute-api.us-east-1.amazonaws.com
Test Profiles	Smoke → Load (sequential)
Monitor	k6 logs: run-k6-profile.sh; CloudWatch: rc-stress-monitor.sh
SLA: API GW 5xx	Must be 0
SLA: Lambda Errors	Must be 0
SLA: DynamoDB Throttles	Must be 0
SLA: API GW Latency p99	< 500ms
SLA: ECS CPU	< 80% sustained
SLA: CloudFront 5xx Rate	< 1%

PageSpeed Context (Pre-Test)

Desktop: 96/100 — PASS. CloudFront + WebP serving correctly. Mobile: 75/100 — below 85 target due to missing **priority** prop on hero image and render-blocking font requests. This is a frontend delivery issue only and does not reflect backend health. TTFB and server-side metrics within SLA.

SLA Gate Results

SLA Gate	Threshold	Observed	Status
Lambda Errors (total)	0	0	PASS
API GW 5xx (60s window)	≤ 1	N/A	—
API GW 4xx (60s window)	≤ 10	N/A	—
API GW Latency p99	< 500ms	N/A	—
ECS CPU Utilization	< 80%	N/A	—
ECS Memory Utilization	< 85%	N/A	—
CloudFront 5xx Rate	< 1%	N/A	—
DynamoDB Throttles	0	0	PASS

k6 Performance Metrics

Smoke Profile

Metric	Value
HTTP Req Duration — p95	77ms
HTTP Request Failure Rate	0.00

Load Profile

Metric	Value
HTTP Req Duration — p95	55ms
HTTP Request Failure Rate	0.00

CloudWatch Metrics Snapshot

Service	Metric	Value	SLA	Status
Lambda	Total Errors	0	0	PASS
API Gateway	5xx Errors (60s)	N/A	≤ 1	—
API Gateway	4xx Errors (60s)	N/A	≤ 10	—
API Gateway	Latency p99	N/A	< 500ms	—
API Gateway	Total Requests	N/A	—	—
DynamoDB	Throttle Events	0	0	PASS
ECS/Fargate	CPU Utilization	N/A	< 80%	—
ECS/Fargate	Memory Utilization	N/A	< 85%	—
CloudFront	5xx Error Rate	N/A	< 1%	—
CloudFront	Total Requests	N/A	—	—

Findings & Recommendations

Frontend — Mobile PageSpeed (Action Required)

Priority	Finding	Fix	Impact
P0	LCP 5.7s on mobile — target <2.5s	Add priority prop to hero <Image> on /product/venue	+8–12 pts
P1	Render-blocking fonts (450ms savings)	Switch to next/font/google or use rel=preload with onload	+3–5 pts
P2	Unused JavaScript (150 KiB)	Dynamic import below-fold components	+2–3 pts
P2	Image sizing gap (260 KiB)	Add correct sizes attr to hero Image for mobile breakpoints	+2–3 pts

Infrastructure — Monitoring (Ongoing)

Area	Recommendation
Lambda	Error rate alarm: > 0 errors in 5-min window → SNS page
DynamoDB	Switch to on-demand capacity for tables hit during stress test
API Gateway	Add CloudWatch dashboard: p99 latency + 5xx rate trending
ECS	Auto-scaling trigger at 70% CPU sustained for 5 min
CloudFront	Enable real-time logging → CloudWatch Insights for pattern detection
Cognito	Throttle alarm: 5 SignInThrottles/5-min → auth storm detection

Next Steps

1. Apply mobile PageSpeed fixes and re-run on /product/venue. Target: ≥ 85.
2. If stress PASS: proceed to pilot traffic onboarding per Rapid Cortex Pilot Offer.
3. If stress FAIL: address each breached SLA gate before any pilot traffic increase.
4. Schedule monthly stress test runs (smoke + load) as part of production ops cadence.
5. Add rc-stress-monitor.sh metrics to a persistent CloudWatch dashboard.